

can't buy them directly from a retail shop near your home or on any of the regular e-commerce web sites but they're available separately and can be purchased by shops from a third-party provider. These processors are designed to be used under many conditions and are usually soldered in place into the final product (there are quite a few steps between selecting a processor and fixing it in the final product).

Microcontrollers

Though not always, many a times an embedded system is designed to control machinery.

In such cases, just a processor is not enough. We also need something which can send signals – such as rotate wheels,

toggle switches, control movements of a device and so on – to other non-computing hardware. For such purposes, we don't use microprocessors, but microcontrollers. Microcontrollers are usually processors with hardware control features. They have a set of instruction codes which, when sent, activate some of their pins that are connected to a mechanical arm controller or a motor directly.

Although they bear significant resemblance to each other, microcontrollers and microprocessors are not the same; microcontrollers can do basic computations and can specialise in some aspects but are not as powerful as microprocessors. Typically the processing capabilities of a microcontroller are less than that of a microprocessor. They're majorly used in places where mechanical movements are needed. When there are no physical arms to be controlled, microcontrollers are mostly not needed.



Intel 8051 is one of the most widely known microcontrollers

System on Chip

In most, if not all cases, the capabilities and usage of embedded systems are pre-determined and the kind of workload it will go through is well known ahead of time. In addition, an embedded system needs to be as compact as possible. These two factors motivate another design known as SoC (System on Chip) wherein most controllers and processors (there's more than just one processor, e.g. signal processor) are built into one chip. This reduces the number of chips on the board, saves space and optimises operations for increased speed. A SoC chip designed for a smartphone would contain